

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Project Information	Details
Pentester Name	Anaam Umar
Program/Batch	B082-Networkwalks
Date	24 August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. networkwalks.com — assigned educational footprinting lab 2. Own/local LAN network
Permission / Scope	Educational and authorized lab activity
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I have performed these activities only within the scope of the assigned educational cybersecurity lab and on my own/local network where applicable. All materials are for education and research purposes only. The tools and techniques demonstrated in this report must not be used against systems, devices or networks without appropriate authorization.

The instructor, authors and Networkwalks are not responsible for misuse of the knowledge contained in this report. Unauthorized access, scanning or security testing may violate laws and organizational policies. Every action performed during practical work should remain within the approved scope.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own/local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they demonstrate how a security professional can move from gathering publicly available information to discovering active hosts on a network.

The practical activities were completed as part of the Week 2 Cybersecurity & Ethical Hacking training program. The footprinting activities were performed in Kali Linux, while Zenmap was used for network discovery. Each activity is documented with the command or procedure used, the observed result, the security significance of the finding, and supporting evidence.

The W2-PM1 practical defines reconnaissance or footprinting as an initial stage in security testing in which information such as domain registration, IP addressing, web technologies, DNS records and WAF presence is collected.

3. Tools Used

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and network scanning activities.
WHOIS	Find domain registration details, registrar information and name servers.
WhatWeb	Fingerprint web technologies, CMS, plugins and related information.
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read HTTP response headers returned by the website.
wafw00f	Detect whether a Web Application Firewall is protecting the target.
DNSRecon	Enumerate DNS information including name servers, mail and service records.
Zenmap (Nmap GUI)	Scan the local subnet and identify live hosts.
Windows CMD	Identify local IP address, subnet and related network configuration.

4. Activities Performed

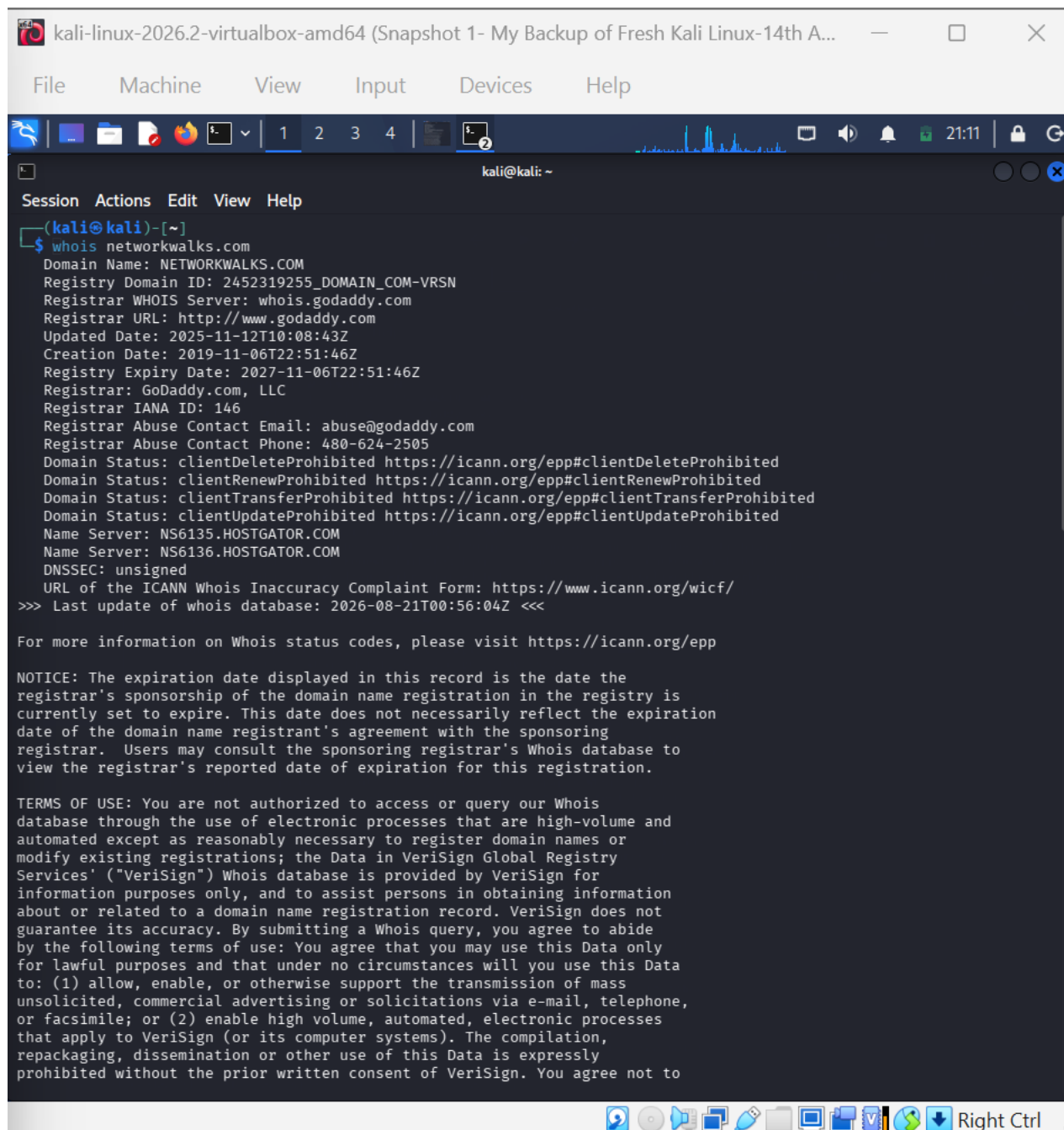
4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target. The practical specifically requires these six tasks.

Task 1 — WHOIS

Command used: `whois networkwalks.com`

WHOIS was used to obtain publicly available domain registration information. The output provides registrar information, registration and expiry information, domain status and name-server details. These details can help a security professional understand the external registration and hosting infrastructure of a domain.



```
kali@kali: ~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-08-21T00:56:04Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for  
information purposes only, and to assist persons in obtaining information  
about or related to a domain name registration record. VeriSign does not  
guarantee its accuracy. By submitting a Whois query, you agree to abide  
by the following terms of use: You agree that you may use this Data only  
for lawful purposes and that under no circumstances will you use this Data  
to: (1) allow, enable, or otherwise support the transmission of mass  
unsolicited, commercial advertising or solicitations via e-mail, telephone,  
or facsimile; or (2) enable high volume, automated, electronic processes  
that apply to VeriSign (or its computer systems). The compilation,  
repackaging, dissemination or other use of this Data is expressly  
prohibited without the prior written consent of VeriSign. You agree not to
```

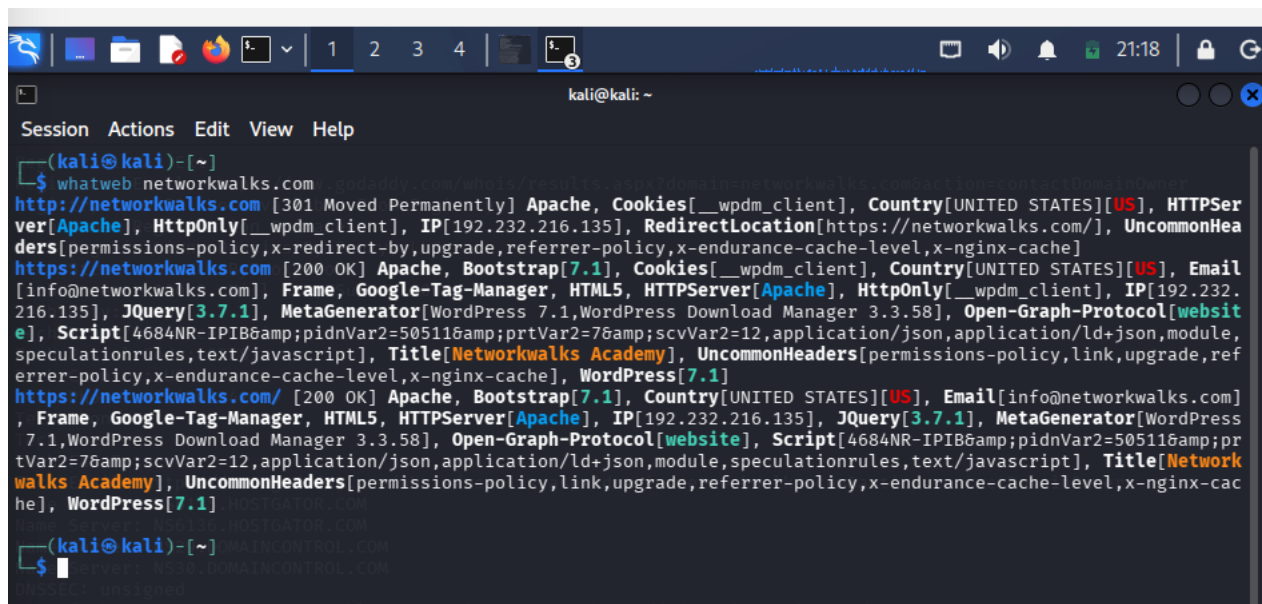
Figure 1: WHOIS command output for networkwalks.com

Security significance: WHOIS information can contribute to an external profile of the target. Name servers and registrar information may provide useful context for later authorized security assessment activities. The practical notes that WHOIS can reveal registration and name-server information.

Task 2 — WhatWeb

Command used: `whatweb networkwalks.com`

WhatWeb was used to fingerprint the technologies running on the website. The observed output identified WordPress 7.0.4 and WP Download Manager 3.3.58, together with other information exposed by the website.



```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

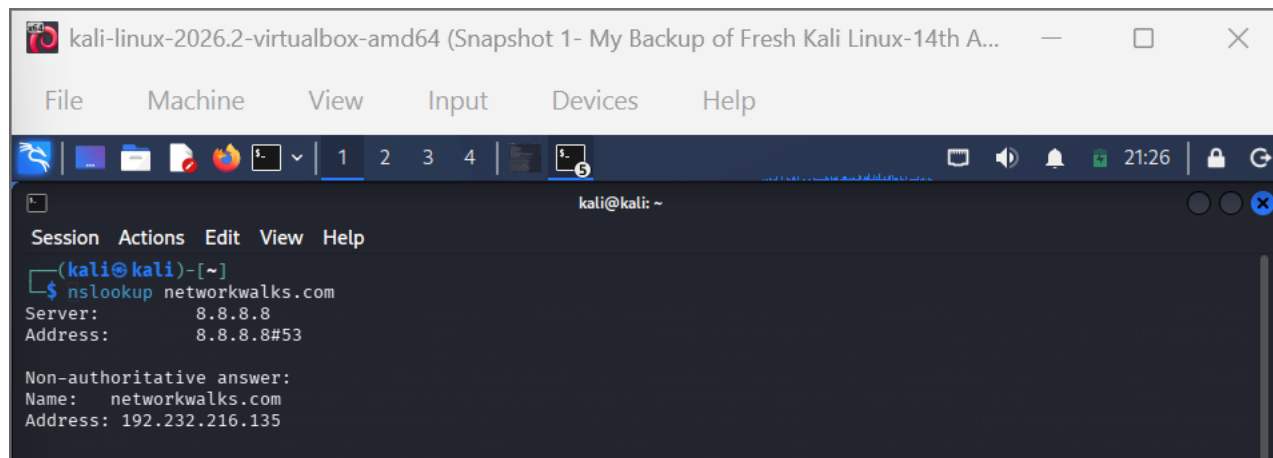
Figure 2: WhatWeb technology fingerprinting output

Security significance: Technology and version information can help identify software that requires security review. The practical explains that exposed versions may be compared with known vulnerability information during an authorized assessment. `filecite``turn3file2``L187-L199`

Task 3 — Nslookup

Command used: `nslookup networkwalks.com`

Nslookup was used to resolve the domain name to its IP address. The observed result identified the address 192.232.216.135.



```
kali-linux-2026.2-virtualbox-amd64 (Snapshot 1- My Backup of Fresh Kali Linux-14th A...
File Machine View Input Devices Help

(kali@kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

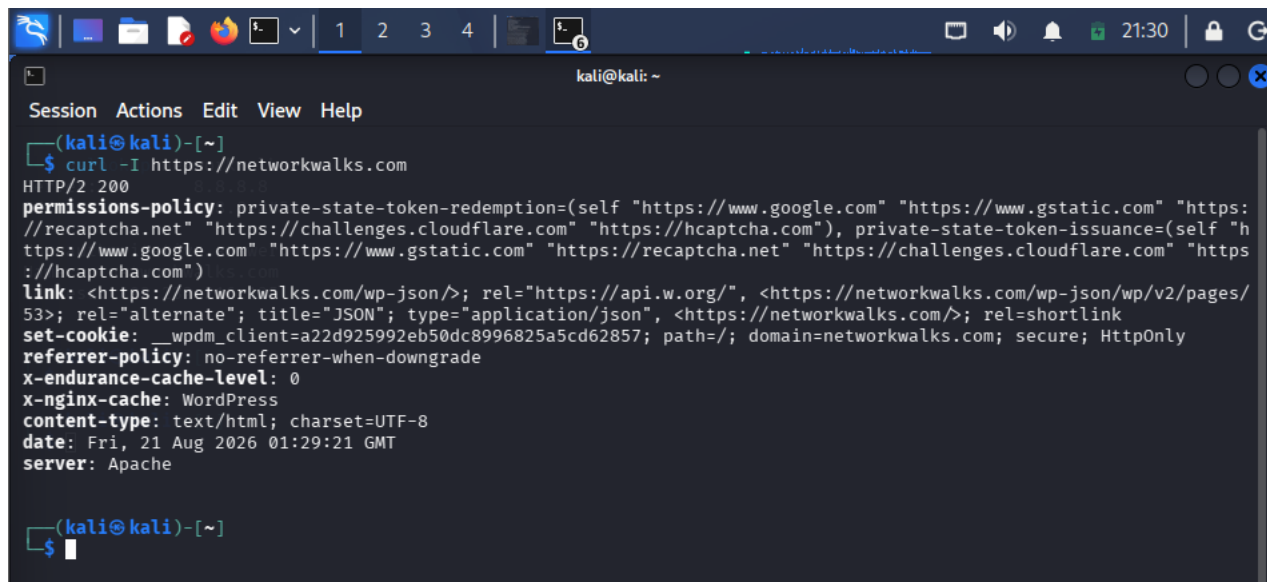
Figure 3: Nslookup resolving networkwalks.com to an IP address

Security significance: Knowing the IP address provides information about the location of the web service and can support further authorized infrastructure mapping.

Task 4 — Curl HTTP Headers

Command used: `curl -I https://networkwalks.com`

Curl was used with the `-I` option to inspect the HTTP response headers. The output provided technical information about the web application and exposed the WordPress REST API endpoint `/wp-json/`.



```
(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=a22d925992eb50dc8996825a5cd62857; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 21 Aug 2026 01:29:21 GMT
server: Apache
```

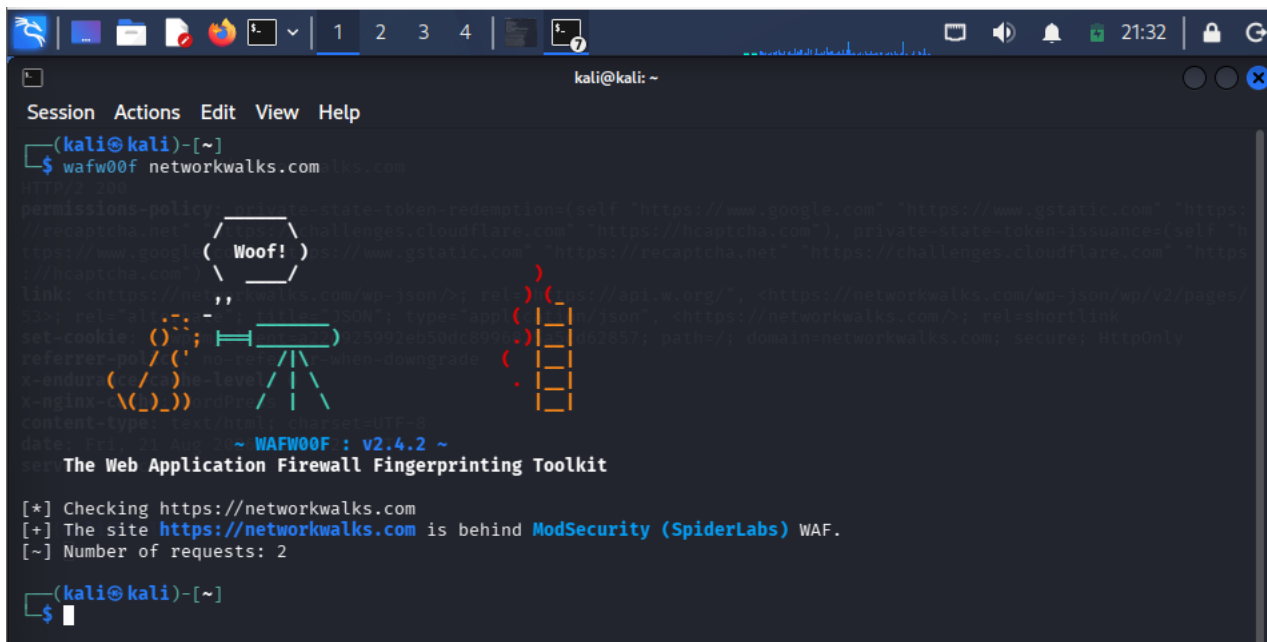
Figure 4: Curl HTTP response header output

Security significance: HTTP headers can expose server, application and caching information that assists technology fingerprinting and further enumeration. The practical specifically identifies the `/wp-json/` endpoint in the observed output.

Task 5 — Wafw00f

Command used: `wafw00f networkwalks.com`

Wafw00f was used to identify whether a Web Application Firewall was protecting the target. The observed result identified ModSecurity (SpiderLabs).



```
(kali@kali)-[~]
$ wafw00f networkwalks.com
Woof!
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=a22d925992eb50dc8996825a5cd62857; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 21 Aug 2026 01:29:21 GMT
server: The Web Application Firewall Fingerprinting Toolkit

[+] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

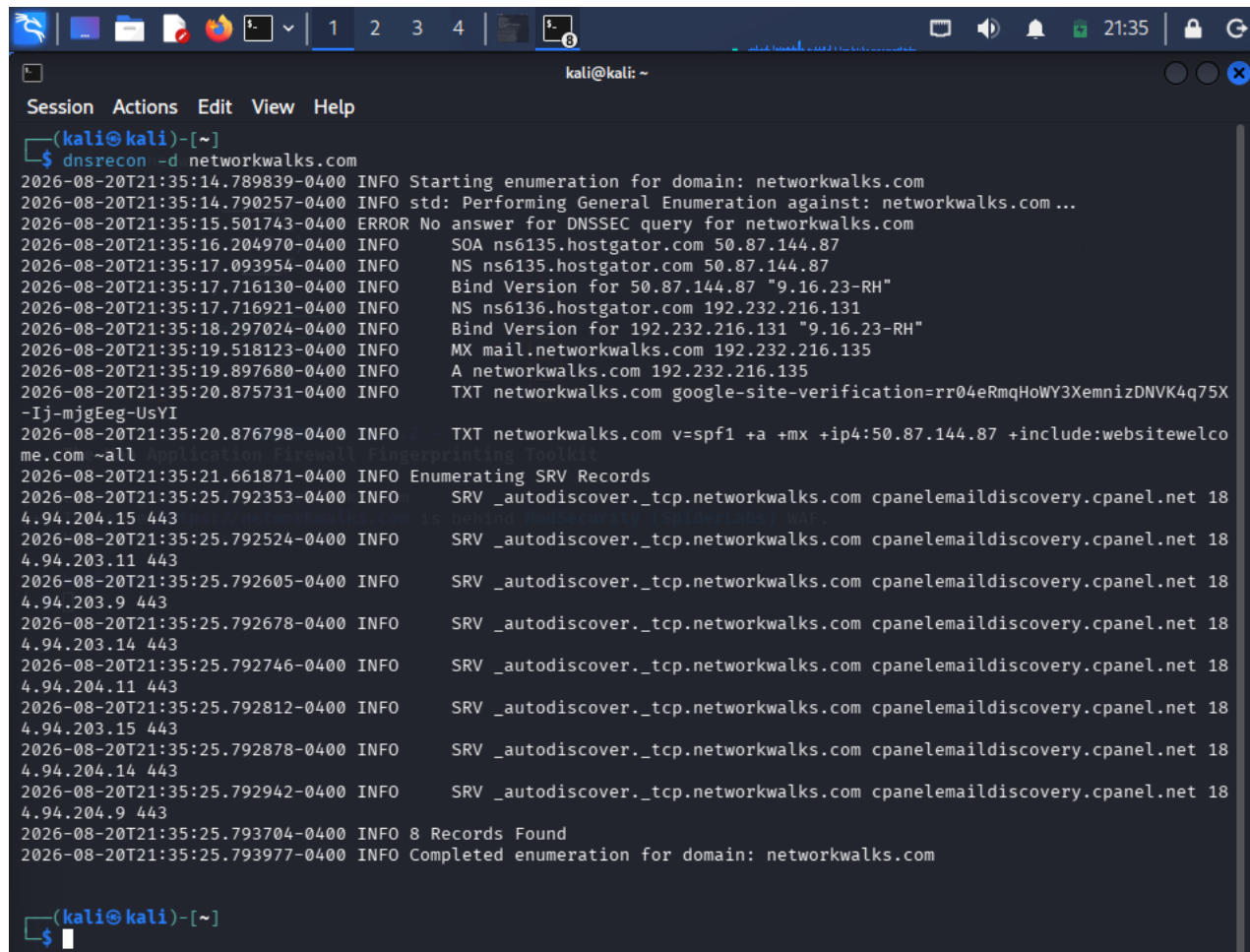
Figure 5: Wafw00f WAF detection output

Security significance: Identifying the presence of a WAF provides useful information about the defensive architecture of a web application. A WAF may block or log certain web requests, so this is relevant when planning authorized testing.

Task 6 — DNSRecon

Command used: `dnsrecon -d networkwalks.com`

DNSRecon was used to enumerate DNS information, including name servers, mail-related records, SPF/TXT information and service records. The practical also notes DNS software and cPanel-related service information in the observed results.



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ dnsrecon -d networkwalks.com  
2026-08-20T21:35:14.789839-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-08-20T21:35:14.790257-0400 INFO std: Performing General Enumeration against: networkwalks.com ...  
2026-08-20T21:35:15.501743-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-08-20T21:35:16.204970-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-08-20T21:35:17.093954-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-08-20T21:35:17.716130-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"  
2026-08-20T21:35:17.716921-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-08-20T21:35:18.297024-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"  
2026-08-20T21:35:19.518123-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-08-20T21:35:19.897680-0400 INFO A networkwalks.com 192.232.216.135  
2026-08-20T21:35:20.875731-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X  
-Ij-mjgEeg-UsYI  
2026-08-20T21:35:20.876798-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelco  
me.com ~all  
2026-08-20T21:35:21.661871-0400 INFO Enumerating SRV Records  
2026-08-20T21:35:25.792353-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.15 443  
2026-08-20T21:35:25.792524-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.11 443  
2026-08-20T21:35:25.792605-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.9 443  
2026-08-20T21:35:25.792678-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.14 443  
2026-08-20T21:35:25.792746-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.11 443  
2026-08-20T21:35:25.792812-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.203.15 443  
2026-08-20T21:35:25.792878-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.14 443  
2026-08-20T21:35:25.792942-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 18  
4.94.204.9 443  
2026-08-20T21:35:25.793704-0400 INFO 8 Records Found  
2026-08-20T21:35:25.793977-0400 INFO Completed enumeration for domain: networkwalks.com  
(kali@kali)-[~]  
$
```

Figure 6: DNSRecon DNS enumeration output

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery on my local network. The practical required identification of the local IP address and LAN subnet, discovery of live hosts, identification of IP and MAC information where available, and generation of a network topology.

The scan was performed against the local subnet 192.168.56.0/24 using the Ping Scan profile. Zenmap displayed the command `nmap -sn 192.168.56.0/24`. The scan identified one live host: 192.168.56.1. The hostname displayed in the Zenmap interface was 192-168-56-1.rev.bb.zain.com.

A MAC address was not displayed in the captured Zenmap results. Therefore, no MAC address has been fabricated or inferred for this report.

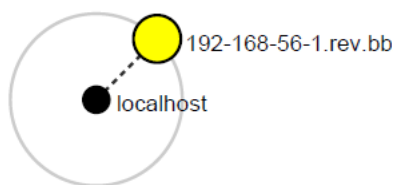


Figure 7: Zenmap Ping Scan and topology result for 192.168.56.0/24

The topology view showed localhost connected to the discovered host 192.168.56.1

4.3 Summary of Results

Activity	Observed Result
WHOIS	Domain registration, registrar and name-server information obtained.
WhatWeb	WordPress 7.0.4 and WP Download Manager 3.3.58 identified.
Nslookup	networkwalks.com resolved to 192.232.216.135.
Curl	HTTP response headers and /wp-json/ information observed.
Wafw00f	ModSecurity (SpiderLabs) identified.
DNSRecon	DNS, mail, SPF/TXT and service-related records enumerated.
Zenmap	192.168.56.0/24 scanned; 1 live host identified: 192.168.56.1.
MAC address	Not displayed in the captured Zenmap result.
Topology	Topology displayed with localhost connected to 192.168.56.1.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, the following potential risks were identified. These are observations from reconnaissance and host discovery, not confirmed vulnerabilities.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager versions.	Exposed technology/version information may help identify software requiring security review.	Medium
2	Server IP address identifiable	Nslookup resolved networkwalks.com to 192.232.216.135.	Provides information about the network location of the web service.	Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/.	May assist technology fingerprinting and further enumeration.	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs).	Reveals information about the web application's defensive architecture.	Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records.	DNS information can help build a broader infrastructure profile.	Medium
6	Live host visible on local network	Zenmap identified one live host at 192.168.56.1.	A live device is visible on the local subnet and should be known and authorized.	Low

Risk level key: Critical | Medium | Low

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules. Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

1. Review publicly exposed technology information

Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.

2. Keep software updated

CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.

3. Review HTTP headers

HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.

4. Review DNS records regularly

DNS records should be checked periodically to ensure that only required information and services are publicly exposed.

5. Properly configure and monitor the WAF

The identified WAF should remain enabled, properly configured and monitored.

6. Perform regular internal network discovery

Organizations should periodically scan their own networks to identify active devices.

7. Investigate unknown devices

Any unexpected device discovered during network scanning should be investigated and verified.

8. Maintain network documentation

Network topology and device information should be documented and updated regularly.

9. Perform security testing with authorization

Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain registration information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. The scan of 192.168.56.0/24 identified one live host, 192.168.56.1, and the topology view was generated. A MAC address was not available in the captured scan output, so it was documented as not displayed.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

Author

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Project Information: Cybersecurity program at Networkwalks | Week 02